

Download two files:

WCHISPTool_Setup.exe : https://www.wch.cn/downloads/WCHISPTool_Setup_exe.html

WCH-LinkUtility.ZIP : https://www.wch.cn/downloads/WCH-LinkUtility_ZIP.html

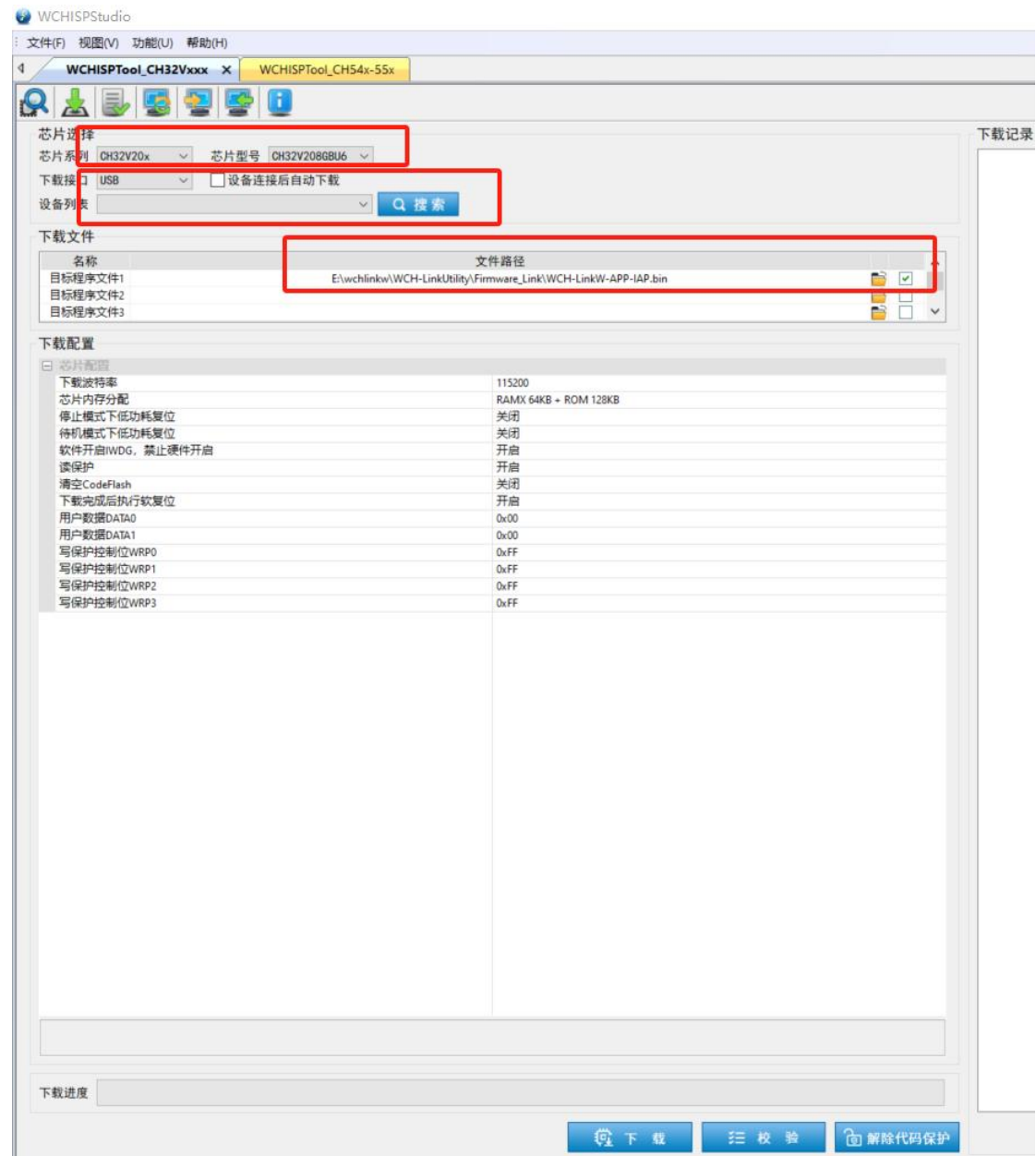
WCH-LinkUtility : Includes drivers, firmware, ----- it is not recommended to use it

for ID setting!!! (Generally, you need to change the ID).

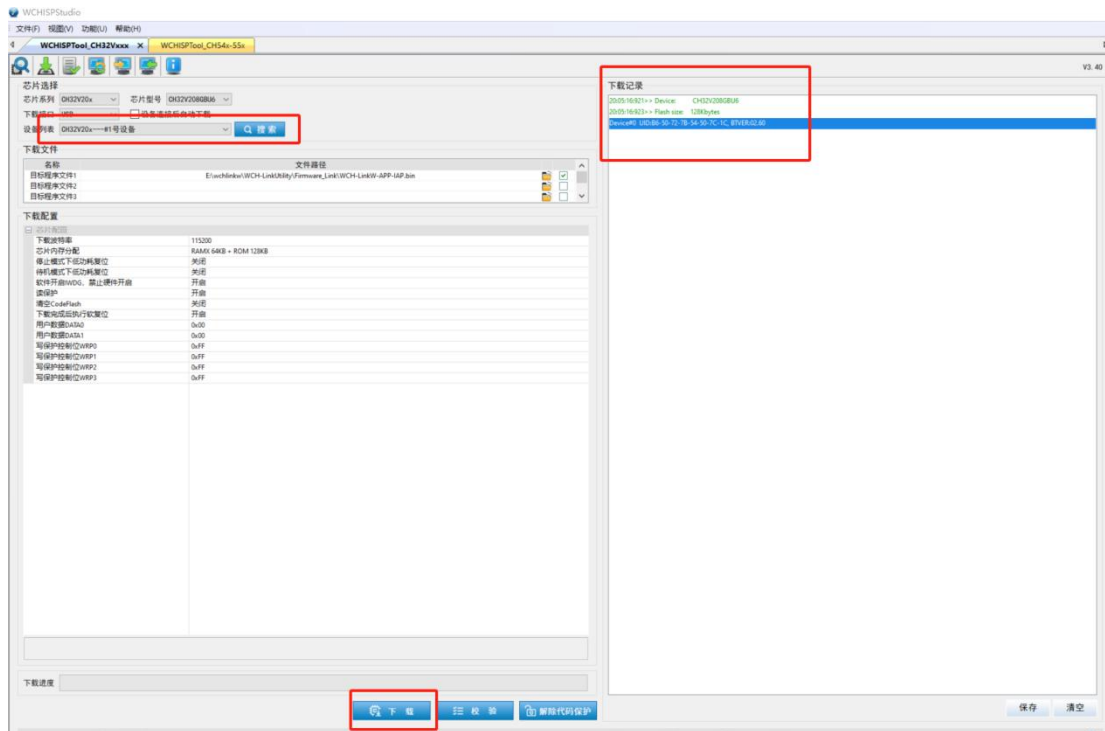
WCHISPTool : Used for firmware download updates.

Firmware Update:

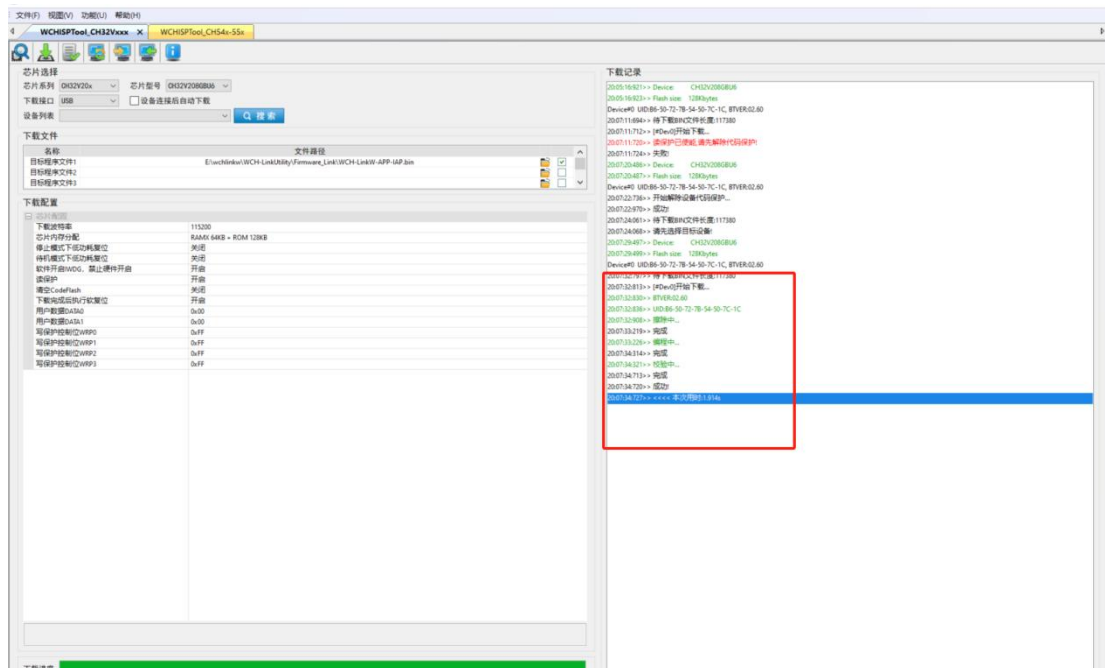
1. Open WCHISPTool_Setup.exe, select ch32v208gbu6 for the chip, and select the latest firmware (located in WCH-LinkUtility\Firmware_Link\WCH-LinkW-APP-IAP.bin).



Take out WCHlinkW, press and hold the boot button to power on until the device list reads the device, and then download. If the download fails, you may need to unprotect the code and download it again.



2、 After the download is complete, the downloader defaults to RV mode



3、 If you want to use arm mode, the downloader is powered off, and press and hold the KEY button to power on. When the downloader blue light is on, switch to ARM mode.

4、 If you're using wireless mode, you'll need to repeat the above for both downloaders. The firmware update is complete

Driver Installation:

Window10:

WCH-LinkUtility.zip open the package, go to the file WCH-LinkUtility\Drv_Link and install the WCHLinkDrv_WHQL_S.exe

Window7:

CDC Device Installation Issues:

- ① If the serial driver is successfully installed, the following steps are not required ②
- ② Confirm whether there is a usbser.sys file in path B, and if it is missing, copy it from path A to path B
- ③ Reinstall the CDC driver (see the above table for the driver path, please install the CDC driver in the corresponding mode)



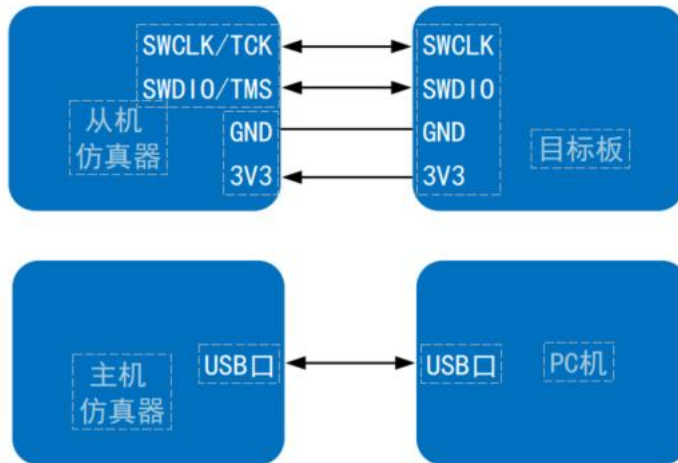
Note: If the above steps do not solve the problem, please refer to the link below



References: http://www.wch.cn/downloads/InstallNoteOn64BitWIN7_ZH_PDF.html

Wireless mode use:

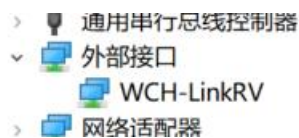
- 1、The slave is connected to the single-chip microcomputer, using 3V3 or 5V (choose one of the two), SWCLK, SWDIO, GND, if you want to use the serial port, connect TX and RX two signal lines. (Note: The MCU system has 5V, and it is recommended to use 5V power supply)



- 2、After the slave connection is completed, the master is connected to the computer. When the green light is on, the two downloaders are connected.

Feedback:

- 1、Open Device Manager, and if there is an exclamation point, install the driver first



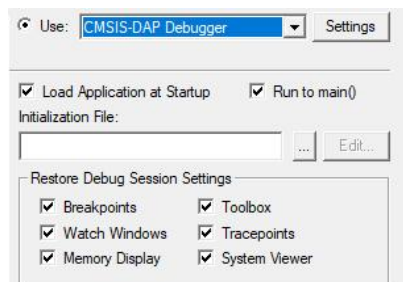
- 2、If the green light flashes after the two downloaders are connected, please update the firmware of the two downloaders on your computer
- 3、If the host is plugged into the computer, the USB device is not read. Please replace a data cable first. If it doesn't work after replacing the data cable, rotate the port of Typec 180° and insert it. (This situation is very rare, half of the Typec port is damaged)

Keil uses:

Note: The keil version must be larger than V525!!

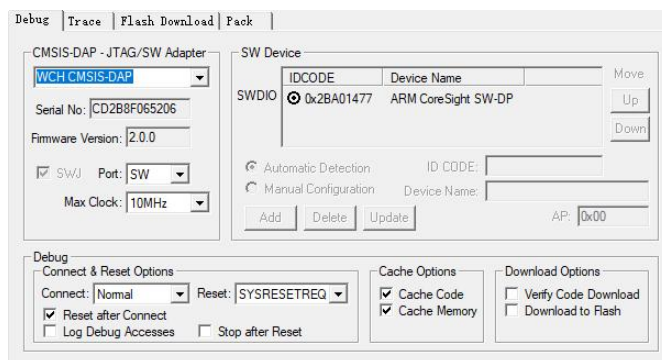
Some keil versions are already larger than the required version, but they still can't be used, please check whether the project is created with the old keil version!!

- ① Click the magic wand in the toolbar to pop up the Options for Target dialog box, click Debug, and select the emulator model



- ② Click the Use box and select CMSIS-DAP Debugger

- ③ Click the Settings button to pop up the Cortex-M Target Driver Setup dialog box



- ④ Click Flash Download to download the configuration

Download Function: 配置选项

- Erase Full Chip: 全片擦除
- Erase Sectors: 部分擦除
- Do not Erase: 不擦除
- Program: 编程
- Verify: 校验
- Reset and Run: 复位后运行

Debug | Trace | Flash Download | Pack |

Download Function

LOAD

☒ Erase Full Chip
☐ Erase Sectors
☐ Do not Erase

☒ Program
☒ Verify
☒ Reset and Run

RAM for Algorithm

Start:

0x20000000

Size:

0x00001000

Programming Algorithm

Description	Device Size	Device Type	Address Range
CH32F1xx Flash	64k	On-chip Flash	08000000H - 0800FFFFH

Start:

Size:

Add

Remove

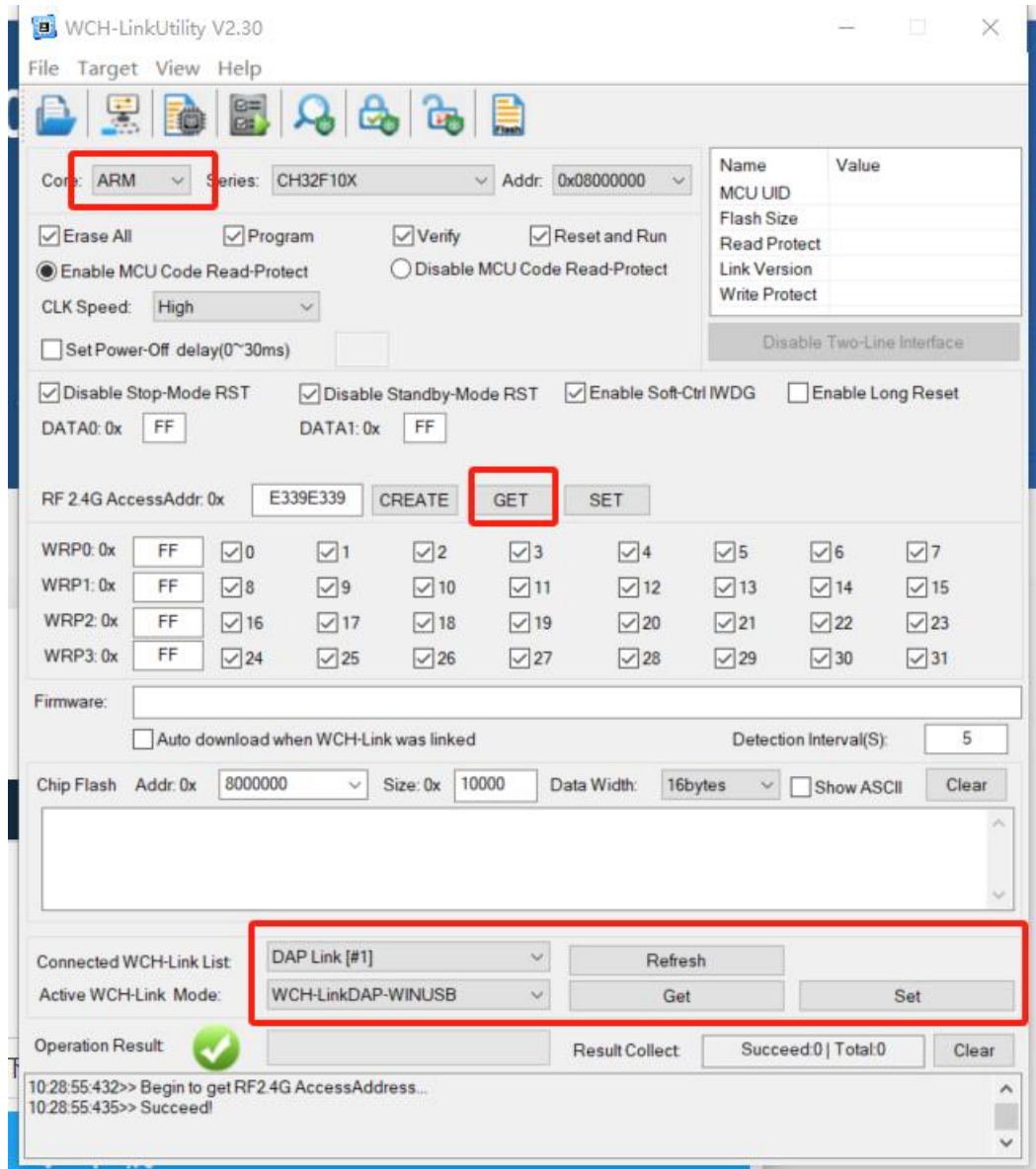
Multiple pairs of wireless simultaneous use:

Note: When multiple pairs of wireless downloaders are used at the same time, use the following method to set the ID to ensure that both downloaders have the same ID.

A. Download :WCH-LinkUtility.zip

Website: https://www.wch.cn/downloads/WCH-LinkUtility_ZIP.html

B. Open the WCH-LinkUtility.exe



- 1、Select ARM mode.
- 2、Conencted WCH-Link List, click Refresh, Active WCH-Link Mode, and then Get.
- 3、RF 2.4G AccessAddr first click GET to get the current downloader ID, which is E339E339 by default.
- 4、RF 2.4G AccessAddrClick CREATE, generate a new ID, click SET, and the setting is complete.
- 5、Perform the same operation on the other downloader to ensure that the IDs of the two

downloaders are the same.